Package 'AURMC'

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Title Area Under the Repeated Measures Curve

Version 0.5.0

Description Estimates and compares areas under curves formed by repeatedly measured outcomes observed subject to censoring and a terminal event. Implements Kaplan-Meier adjustment for censoring, influence-function standard errors, Wald inference, and multiplier perturbation inference, with utilities for interpolation, simulation, tabulation, and plotting. For additional details, see: McCaw ZR, Armbruster S, Tian L, Claggett BL, Fitzmaurice G, Foà A, Pfeffer MA, Wei LJ (2026) <doi:10.1016/j.jchf.2026.103013>.

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`AtRiskMatrixR` *Tabulate At-Risk Matrix R*

Description

Construct a subject (row) by evaluation time (col) matrix.

Usage

```
AtRiskMatrixR(eval_times, idx, time)
```

Arguments

`eval_times`	Evaluation times.
`idx`	Unique subject index.
`time`	Observation time.

Value

Numeric matrix.

AURMC *Area Under the Repeated Measures Curve*

Description

Area Under the Repeated Measures Curve

Usage

```
AURMC(
  data,
  alpha = 0.05,
  censor_after_last = TRUE,
  idx_name = "idx",
  int_method = "trapezoid",
  perturbations = NULL,
  random_state = 0,
  status_name = "status",
  tau = NULL,
  time_name = "time",
  value_name = "value"
)
```

Arguments

data	Data.frame.
alpha	Type I error.
censor_after_last	Introduce censoring after the last event *if* no observation-terminating event is present.
idx_name	Name of column containing a unique subject index.
int_method	Integration method, selected from "left", "right", "trapezoid".
perturbations	Number of perturbations to use for bootstrap inference. If NULL, only analytical inference is performed.
random_state	Seed to ensure perturbations are reproducible.
status_name	Name of column containing the status. Must be coded as 0 for censoring, 1 for a measurement, 2 for death. Each subject should have an observation-terminating event, either censoring or death.
tau	Truncation time.
time_name	Name of column containing the observation time.
value_name	Name of the column containing the measurement.

Value

A data.frame containing the method, truncation time, area estimate, standard error, confidence limits, and p-value.

Examples

```
example_data <- data.frame(
  idx = rep(1:3, each = 2),
  time = rep(c(0, 1), 3),
  status = rep(c(1, 0), 3),
  value = c(1, 1, 2, 2, 3, 3)
)
AURMC(example_data, tau = 1)
```

AURMC-class *AURMC Object*

Description

AURMC Object

Slots

`Arm0` Results for the reference arm.

`Arm1` Results for the treatment arm.

`Contrast` Contrasts.

BootstrapSamplesR *Bootstrap Samples R*

Description

Returns a bootstrap (row) by statistic (col) matrix. If `return_auc = TRUE`, the output has a single column, the AUC. If `return_auc = FALSE`, the output has a single column for each evaluation time.

Usage

```
BootstrapSamplesR(
  boot,
  eval_times,
  idx,
  status,
  time,
  value,
  int_method = "trapezoid",
```

```
  replace_na = FALSE,
  return_auc = FALSE,
  trunc_time = NULL
)
```

Arguments

boot	Bootstrap replicates.
eval_times	Evaluation times.
idx	Unique subject index.
status	Status, coded as 0 for censoring, 1 for a measurement, 2 for a terminal event.
time	Observation time.
value	Observation value.
int_method	Integration method, selected from "left", "right", "trapezoid".
replace_na	Replace NaN with zero?
return_auc	Return the AUC?
trunc_time	Truncation time? Optional. If omitted, defaults to the maximum evaluation time.

Value

Numeric matrix.

CalcMartingaleR *Calculate Martingale*

Description

Calculate $dM_i(t) = dN_i(t) - Y_i(t)d\Lambda(t)$.

Usage

```
CalcMartingaleR(haz, idx, status, time, unique_times)
```

Arguments

haz	Value of the hazard at each unique time.
idx	Subject index.
status	Status, coded as 0 for censoring, 1 for a measurement, 2 for a terminal event.
time	Subject observation times.
unique_times	Unique times at which to obtain the martingale.

Value

Matrix with subjects as rows and unique times as columns.

CalcMuR *Calculate Mu R*

Description

Evaluate the quadrature tail associated with $\mu(t;\tau) = \int_t^\tau S(u-)d(u)/y(u)du$. The current grid point is excluded because the hazard jump at t affects $\hat{S}(u-)$ only for $u > t$.

Usage

```
CalcMuR(d, surv, unique_times, y, int_method = "trapezoid")
```

Arguments

d	Value of d(t) at each time point.
surv	Value of $S(t-)$ at each time point.
unique_times	Unique values of time t.
y	Value of y(t) at each time point.
int_method	Integration method, selected from "left", "right", "trapezoid".

Value

Numeric vector of $\mu(t;tau)$.

CensorAfterLast *Censor After Last*

Description

Introduce a censoring after the last event if no observation terminating event is present.

Usage

```
CensorAfterLast(data)
```

Arguments

data	Data.frame.

Value

None.

CompareAURMCs *Compare Areas Under Repeated Measures Curves*

Description

Compare Areas Under Repeated Measures Curves

Usage

```
CompareAURMCs(
  data,
  alpha = 0.05,
  arm_name = "arm",
  censor_after_last = TRUE,
  idx_name = "idx",
  int_method = "trapezoid",
  perturbations = NULL,
  random_state = 0,
  status_name = "status",
  tau = NULL,
  time_name = "time",
  value_name = "value"
)
```

Arguments

data	Data.frame.
alpha	Type I error.
arm_name	Name of the column containing treatment arm. Coded as 0 for reference, 1 for treatment.
censor_after_last	Introduce censoring after the last event *if* no observation-terminating event is present.
idx_name	Name of column containing a unique subject index.
int_method	Integration method, selected from "left", "right", "trapezoid".
perturbations	Number of perturbations to use for bootstrap inference. If NULL, only analytical inference is performed.
random_state	Seed to ensure perturbations are reproducible.
status_name	Name of column containing the status. Must be coded as 0 for censoring, 1 for a measurement, 2 for death. Each subject should have an observation-terminating event, either censoring or death.
tau	Truncation time.
time_name	Name of column containing the observation time.
value_name	Name of the column containing the measurement.

Value

An object of class AURMC containing arm-specific estimates and difference and ratio contrasts. Ratio inference is returned as NA when either arm-specific area is non-positive.

Examples

```
example_data <- data.frame(
  idx = rep(1:4, each = 2),
  arm = rep(c(0, 0, 1, 1), each = 2),
  time = rep(c(0, 1), 4),
  status = rep(c(1, 0), 4),
  value = rep(c(1, 1, 2, 2), each = 2)
)
CompareAURMCs(example_data, tau = 1)
```

Curve *Estimate Curve*

Description

Estimate Curve

Usage

```
Curve(
  data,
  censor_after_last = TRUE,
  idx_name = "idx",
  status_name = "status",
  time_name = "time",
  value_name = "value"
)
```

Arguments

data	Data.frame.
censor_after_last	Introduce censoring after the last event *if* no observation-terminating event is present.
idx_name	Name of column containing a unique subject index.
status_name	Name of column containing the status. Must be coded as 0 for censoring, 1 for a measurement, 2 for death. Each subject should have an observation-terminating event, either censoring or death.
time_name	Name of column containing the observation time.
value_name	Name of the column containing the measurement.

Value

Function.

`DrawBootstrapR` *Draw Bootstrap R*

Description

Draw Bootstrap R

Usage

```
DrawBootstrapR(idx, status, time, value)
```

Arguments

`idx`	Unique subject index.
`status`	Status, coded as 0 for censoring, 1 for a measurement, 2 for a terminal event.
`time`	Observation time.
`value`	Observation value.

Value

Numeric matrix.

`EstimatorR` *Tabulate Estimator R*

Description

Tabulate Estimator R

Usage

```
EstimatorR(
  idx,
  status,
  time,
  value,
  eval_times = NULL,
  int_method = "trapezoid",
  replace_na = FALSE,
  return_auc = FALSE,
  trunc_time = NULL
)
```

Arguments

idx	Unique subject index.
status	Status, coded as 0 for censoring, 1 for a measurement, 2 for a terminal event.
time	Observation time.
value	Observation value.
eval_times	Evaluation times. If omitted, defaults to the unique values of time.
int_method	Integration method, selected from "left", "right", "trapezoid".
replace_na	Replace NaN with zero? Default: FALSE.
return_auc	Return the AUC? Default: FALSE.
trunc_time	Truncation time? Optional. If omitted, defaults to the maximum evaluation time.

Value

Data.frame.

GenData *Simulate Repeated Measures Data*

Description

Simulates repeated measures data at regular time points subject to exponential censoring and death.

Usage

```
GenData(
  censoring_rate,
  death_rate,
  n,
  tau,
  last_missing = FALSE,
  value_mean = 0,
  value_sd = 1
)
```

Arguments

censoring_rate	Rate for the time to censoring.
death_rate	Rate for the time to death.
n	Number of subjects.
tau	Truncation time.
last_missing	Should the last value be missing? Default: FALSE.
value_mean	Mean of measurement.
value_sd	Standard deviation of measurement.

Value

A data.frame with subject index, observation time, status, and repeated measurement value.

Examples

```
set.seed(1)
GenData(censoring_rate = 0.5, death_rate = 0.25, n = 5, tau = 2)
```

`InfluenceR` *Influence Function R*

Description

Influence function contributions for the AUC. Includes the three component integrals (`i1`, `i2`, `i3`) and the overall influence `psi`.

Usage

```
InfluenceR(idx, status, time, trunc_time, value, int_method = "trapezoid")
```

Arguments

`idx`	Unique subject index.
`status`	Status, coded as 0 for censoring, 1 for a measurement, 2 for a terminal event.
`time`	Observation time.
`trunc_time`	Truncation time? Optional. If omitted, defaults to the maximum evaluation time.
`value`	Observation value.
`int_method`	Integration method, selected from "left", "right", "trapezoid".

Value

Data.frame.

`Interpolate` *Interpolate*

Description

Linearly interpolates between each subject's measurements. The input data should contain no missing values.

Usage

```
Interpolate(
  data,
  grid,
  idx_name = "idx",
  status_name = "status",
  rm_na = TRUE,
  time_name = "time",
  value_name = "value"
)
```

Arguments

`data`	Data.frame.
`grid`	Grid of unique points at which to interpolate.
`idx_name`	Name of column containing a unique subject index.
`status_name`	Name of column containing the status. Must be coded as 0 for censoring, 1 for a measurement, 2 for death. Each subject should have an observation-terminating event, either censoring or death.
`rm_na`	Remove records interpolated to NA?
`time_name`	Name of column containing the observation time.
`value_name`	Name of the column containing the measurement.

Value

A data.frame containing subject index, status, time, and the interpolated value.

Notes

The grid of points will be augmented to include the time of each subjects observation terminating event.

Examples

```
example_data <- data.frame(
  idx = c(1, 1, 2, 2),
  status = c(1, 0, 1, 2),
  time = c(0, 1, 0, 1),
  value = c(0, 1, 0, -1)
)
Interpolate(example_data, grid = c(0, 0.5, 1))
```

`InterpolateR` *Interpolation R Linearly interpolates between each subject's measurements. The input data should contain no missing values.*

Description

Interpolation R

Linearly interpolates between each subject's measurements. The input data should contain no missing values.

Usage

```
InterpolateR(grid, idx, status, time, value)
```

Arguments

`grid`	Grid of unique points at which to interpolate.
`idx`	Unique subject index.
`status`	Status, coded as 0 for censoring, 1 for a measurement, 2 for a terminal event.
`time`	Observation time.
`value`	Observation value.

Value

Data.frame.

`KaplanMeierR` *Tabulate Kaplan Meier R*

Description

Constructs a matrix with evaluation times as rows, and 4 columns:

- time Evaluation times.
- nar Number at risk.
- surv Left-limit survival probability, $\hat{S}(t-)$.
- haz Hazard.

Usage

```
KaplanMeierR(eval_times, idx, status, time)
```

Arguments

`eval_times`	Evaluation times.
`idx`	Unique subject index.
`status`	Status, coded as 0 for censoring, 1 for a measurement, 2 for a terminal event.
`time`	Observation time.

Value

Data.frame.

`PerturbationR` *Perturbation R Generates realizations of* $\frac{1}{n}\sum_i = 1\hat{}n\psi_iw_i$, *where* ψ_i *is the influence function for the ith subject and the* w_i *are IID random weights.*

Description

Perturbation R

Generates realizations of $\frac{1}{n}\sum_{i=1}^{n}\psi_i w_i$, where ψ_i is the influence function for the ith subject and the w_i are IID random weights.

Usage

```
PerturbationR(
  idx,
  perturbations,
  status,
  time,
  trunc_time,
  value,
  int_method = "trapezoid"
)
```

Arguments

idx	Unique subject index.
perturbations	Number of perturbations
status	Status, coded as 0 for censoring, 1 for a measurement, 2 for a terminal event.
time	Observation time.
trunc_time	Truncation time? Optional. If omitted, defaults to the maximum evaluation time.
value	Observation value.
int_method	Integration method, selected from "left", "right", "trapezoid".

Value

Numeric vector.

Notes

The random seed should be set in R prior to calling this function.

PlotTwoSample *Plot Two Sample*

Description

Plot Two Sample

Usage

```
PlotTwoSample(
  data,
  arm_name = "arm",
  censor_after_last = TRUE,
  color_labs = c("Ctrl", "Trt"),
  colors = c("#EFC000FF", "#6385B8"),
```

```
  eval_pts = 200,
  idx_name = "idx",
  show_auc = FALSE,
  show_nar = TRUE,
  status_name = "status",
  tau = NULL,
  time_name = "time",
  title = NULL,
  value_name = "value",
  x_breaks = NULL,
  x_labs = NULL,
  x_name = "Time",
  x_max = NULL,
  y_name = "Expected Value",
  y_lim = NULL
)
```

Arguments

data	Data.frame.
arm_name	Name of the column containing treatment arm. Coded as 0 for reference, 1 for treatment.
censor_after_last	Introduce censoring after the last event *if* no observation-terminating event is present.
color_labs	Color labels.
colors	Colors for control and treatment respectively.
eval_pts	Evaluation points between 0 and tau.
idx_name	Name of column containing a unique subject index.
show_auc	Show area under the curve?
show_nar	Show number at risk?
status_name	Name of column containing the status. Must be coded as 0 for censoring, 1 for a measurement, 2 for death. Each subject should have an observation-terminating event, either censoring or death.
tau	Truncation time.
time_name	Name of column containing the observation time.
title	Plot title.
value_name	Name of the column containing the measurement.
x_breaks	X-axis breaks.
x_labs	X-axis labels.
x_name	X-axis name.
x_max	X-axis upper limit.
y_name	Y-axis name.
y_lim	Y-axis limits.

Value

ggplot

---

print.AURMC *Print Method for AURMC Object*

Description

Print method for objects of class AURMC.

Usage

```
## S3 method for class 'AURMC'
print(x, ...)
```

Arguments

x	An object of class AURMC.
...	Unused.

Value

Invisibly returns NULL. This method is called for its side effect of displaying the reference-arm, treatment-arm, and contrast tables stored in x, with noninteger numeric values rounded to three significant digits.

---

show,AURMC-method *Show Method for AURMC Object*

Description

Show Method for AURMC Object

Usage

```
## S4 method for signature 'AURMC'
show(object)
```

Arguments

object	An object of class AURMC.

TabRMC *Tabulate the Repeated Measures Curve*

Description

Tabulate the Repeated Measures Curve

Usage

```
TabRMC(
  data,
  censor_after_last = TRUE,
  idx_name = "idx",
  status_name = "status",
  tau = NULL,
  time_name = "time",
  value_name = "value"
)
```

Arguments

data	Data.frame.
censor_after_last	Introduce censoring after the last event *if* no observation-terminating event is present.
idx_name	Name of column containing a unique subject index.
status_name	Name of column containing the status. Must be coded as 0 for censoring, 1 for a measurement, 2 for death. Each subject should have an observation-terminating event, either censoring or death.
tau	Truncation time.
time_name	Name of column containing the observation time.
value_name	Name of the column containing the measurement.

Value

A data.frame tabulating time, number and proportion at risk, terminal-event hazard, left-limit survival $\hat{S}(t-)$, observed-value mean, and the estimated repeated measures curve.

`ValueMatrixR` *Tabulate Value Matrix R*

Description

Tabulate $D_i(t)$ with subjects as rows and evaluation times as columns.

Usage

```
ValueMatrixR(eval_times, idx, time, value)
```

Arguments

`eval_times`	Evaluation times.
`idx`	Unique subject index.
`time`	Observation time.
`value`	Observation value.

Value

Numeric matrix.

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